

# Code Layout, Readability and Reusability - OptiCrop

| S.No | Code Quality Parameter    | Followed | Remarks                                                    |
|------|---------------------------|----------|------------------------------------------------------------|
| 1    | Consistent Indentation    | Yes      | 4-space indentation used                                   |
| 2    | Proper File Structure     | Yes      | Code organized into templates, static, model and app files |
| 3    | Meaningful Variable Names | Yes      | Clear variable names for soil, crop and prediction         |
| 4    | Function/Method Names     | Yes      | Descriptive function names                                 |
| 5    | Code Comments             | Yes      | Important logic documented                                 |
| 6    | Modular Design            | Yes      | Separate training and prediction modules                   |
| 7    | No Redundant Code         | Yes      | Unused code removed                                        |
| 8    | Error Handling            | Yes      | Input validation and exception handling included           |

## Reusable Components

| Component           | Technology   | Where Reused          | Level  |
|---------------------|--------------|-----------------------|--------|
| Flask App           | Python/Flask | Prediction API        | High   |
| ML Model            | Scikit-learn | Crop Prediction       | High   |
| Data Preprocessing  | Pandas       | Training & Prediction | High   |
| HTML Templates      | HTML/CSS     | User Interface        | Medium |
| Prediction Function | Python       | Multiple Requests     | High   |

## Overall Assessment

| Aspect                  | Rating | Comments               |
|-------------------------|--------|------------------------|
| Code Layout & Structure | 5/5    | Well organized         |
| Readability             | 5/5    | Easy to understand     |
| Reusability             | 5/5    | Reusable modules       |
| Documentation           | 4/5    | Sufficient comments    |
| Overall Score           | 19/20  | Excellent code quality |